ample table: Customer

customer_id	cust_name	city	grade	salesman_id
3002	Nick Rimando	New York	100	5001
3007	Brad Davis	New York	200	5001
3005	Graham Zusi	California	200	5002
3008	Julian Green	London	300	5002

Sample table: salesman

salesman_id	name	city	commission
5001	James Hoog	New York	0.15
5002	Nail Knite	Paris	0.13
5005	Pit Alex	London	0.11
5006	Mc Lyon	Paris	0.14

Sample table: orders

ord_no	purch_amt	ord_date	customer_id	salesman_id
70001	150.5	2012-10-05	3005	5002
70009	270.65	2012-09-10	3001	5005
70002	65.26	2012-10-05	3002	5001
70004	110.5	2012-08-17	3009	5003

1. write a SQL query to find the salesperson and customer who reside in the same city. Return Salesman, cust_name and city.
2. write a SQL query to find salespeople who received commissions of more than 12 percent from the company. Return Customer Name, customer city, Salesman name, commission.
3. write a SQL query to find those customers with a grade less than 300. Return cust_name, customer city, grade, Salesman, salesmancity. The result should be ordered by ascending customer_id.
4. write a SQL query to find those orders where the order amount exists between 500 and 2000. Return ord_no, purch_amt, cust_name, city.
5. write a SQL query to find those orders those are placed in October and November. Return cust_name, salesman_name and order_date.
6. write a SQL query to find the customers those are not placed any order. Return cust_name, salesman_name.

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Semester Final Examination 2023

Level 3 Semester I Course Code: CSE 303 Credit: 3.0

Course Title: Database

Duration: 3 Hours **Total Marks: 90**

[NB: The figure in the right margin indicates the marks for the respective question and split answer of any question is unacceptable]

Section-A

Answer any 3 (three) questions from the following

1. a) Define: Data, Database, Database-management system. 6
   b) Briefly explain about DDL and DML with example. 5
   c) Describe integrity constraints with example. 4

2. a) Explain transitive dependency and multi-valued dependency with example. 8
   b) Briefly explain the level of abstraction in database management system with figure. 7

3. a) DOCTOR Table: 12

id	name	dept	gender	experience
101	MARRY	ENT	M	12
104	SMITH	ORTHOPEDIC	M	5
107	GEORGE	CARDIOLOGY	M	10
114	LARA	SKIN	F	3
109	K GEORGE	MEDICINE	F	9
105	JOHNSON	ORTHOPEDIC	M	10
117	LUCY	ENT	F	3
111	BILL	MEDICINE	F	12
130	MORPHY	ORTHOPEDIC	M	15

SALARY Table:

id	basic	allowances	consultation
101	12000	1000	300
104	23000	2300	500
107	32000	4000	500
114	12000	5200	100
109	42000	1700	200
105	18900	1690	300
130	21700	2600	300

Write the SQL statements for the following:

i. Display name of all doctors who are in "MEDICINE" having more than 10 years' experience and basic salary more than 10000.
ii. Display the average of all the doctors working in "ENT" department using the table DOCTOR and (SALARY.salary = BASIC + ALLOWANCES)
iii. Display the highest consultation fee among all male doctors.
iv. Display the name and basic salary of all the doctors having less than 5 years' experience.

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Semester Final Examination 2024

Level 4 Semester I Course Code: CSE 409 Credit: 3.0

Course Title: Advanced Database

Duration: 3 Hours **Total Marks: 90**

[NB: The figure in the right margin indicates the marks for the respective question and split answer of any question is unacceptable]

Section-A

Answer any 3 (three) questions from the following

1. a) What are the limitations of traditional DBMS that advanced database overcome? Give two examples for each of traditional DBMS and advanced database applications. 3+4

   b) Compare relational database systems with object-oriented databases in terms of data model, schema design, and query language. 8

2. a) Define the term "complex data" in the context of advanced databases. 3

   b) Assume a table student where each student has a complex data type called department, which contains the department's name, student_id, and course (course_code and course_title). Find all students from the CSE department taking the course named 'Database'. 5

   c) A company needs to store and manage images, videos, sensor data, and regular text data together. Can a traditional DBMS handle this requirement? If not, recommend a suitable advanced database solution and explain how it meets the needs that a traditional DBMS cannot. 7

3. a) Customer table: 10

Customer_ID	Name	City
1	Mike	New York
2	Alice	Los Angeles
3	Sara	Chicago
4	Jane	New York

Orders Table:

Order_ID	Customer ID	Order Date	Order total
100	1	2025-07-01	100.00
101	2	2025-06-15	50.00
102	3	2025-07-05	150.00
103	1	2025-07-07	75.00
104	4	2025-07-02	200.00

Write the SQL statements for the following scenarios:

i. Find the total number of orders placed by each customer, excluding orders placed in June.
ii. Find the customer who has placed the highest total order value.
iii. List all orders placed on specific dates (eg., 2025-07-04 and 2025-07-06) and their corresponding customer names.
iv. Find the average order value for each city.
v. Identify customers who haven't placed any orders.

b) Define Online Analytical Processing (OLAP). Discuss the key features of it. 5

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CSE 304
Database Sessional

Quiz: (Marks 7.5)

1. Consider the following relational schema. An employee can work in more than one department:
   Emp (E_id, E_name, Salary)
   Dept (d_id, d_name, manager_id, floor_number)
   Queries:
   i. Print all the names of all employees, who work on the 10th floor and earn salary less than Rs.50,000.
   ii. Print the names of the departments that employee Santa work in.
   iii. Give every employee who works in the toys dept. at 10% raise in the salary.

Mid: Marks (15)

2. Consider the following tables which give details of customers, trucks and packets booked by customers, which are carried by trucks and write SQL commands to do the following:
   Tables:
   CUSTOMER (c_no, c_name, c_address)
   TRUCK (t_no, driver_name)
   PACKET (p_no, c_no, t_no, date_of_booking, weight, destination)
   Queries:
   i. Destinations which have received more than 10 packets.
   ii. Name of customers who have sent at least one packet of weight more than one kg to 'BOMBAY'.
   iii. Name of all customers whose packets were delivered a driver whose name is 'RAJA'
   iv. Top three customers name in terms of total packet weight sent by them. (descending order of total weight)
   v. Name of all customers whose individual shipments are less than one kg.

Final: (Marks 30)

3. **EmployeeInfo Table:**

1	Sanjay	Mehra	HR	P1	Hyderabad(HYD)	01/12/1976	M
2	Ananya	Mishra	Admin	P2	Delhi(DEL)	02/05/1968	F
3	Rohan	Diwan	Account	P3	Mumbai(BOM)	01/01/1980	M
4	Sonia	Kulkarni	HR	P1	Hyderabad(HYD)	02/05/1992	F
5	Ankit	Kapoor	Admin	P2	Delhi(DEL)	03/07/1994	M

EmployeePosition Table:

1	Manager	01/03/2022	500000
2	Executive	02/05/2022	75000
3	Manager	01/04/2022	90000
2	Lead	02/06/2022	85000
1	Executive	01/09/2022	300000

   i. Print the date of joining for all the employees whose name start with R.
   ii. Print all the employee position hold by female gender.
   iii. Print the average salary of male employees.
   iv. Print the project names those are conducted from Delhi.
   v. Print the highest salary among all departments.
   vi. Print the name, department and employee position from the above table.
   vii. Print all the employee names whose joining date in between March to June.
   viii. Print the names and joining dates whose date of birth before 1975.
   ix. Print all the last names of male those are manager.
   x. Print the names and genders those are in Lead position.